Locked decisions — Month 1

Updated as we proceed. Change only deliberately.

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Decision	Choice	Why
Title	Auditable AI Agents: An Immutable On-Chain Audit Trail for LLM Agent Actions	Clear, demo-friendly, MEng-scoped
Core claim	Any modification of an anchored log (edit, delete, insert, reorder) is detected and located	Provable by test; does not depend on model quality
What goes on-chain	Merkle roots only, never raw logs	Cost and privacy
Hash function	SHA-256	In Python hashlib and in Solidity sha256(), so Python and the contract agree
Entry serialisation	Canonical JSON (sorted keys, no spaces, UTF-8)	Same bytes on every machine
Entry linking	Each entry hashes prev_hash + payload	Detects deletion and reordering
Merkle tree	Domain-separated hashes (0x00 leaf, 0x01 node); an unpaired node is carried up unchanged	Avoids second-preimage and duplicate-leaf issues
Smart contract language	Solidity	Standard, well documented
Chain	Local development chain (Hardhat); public testnet only if time allows	Free, fast, fake money
Python	Pure functions, frozen dataclasses, no shared mutable state	Aligns with paradigm-checker skill
Demo agent	Simple multi-step tool-using LLM agent	Agent quality is not the object of study
Demo UI	Streamlit	Month 4
Code style	Prefer pure functions / immutable data	Easier to test and reason about

Research questions (locked for proposal)

1. Does the verifier detect 100% of edit, delete, insert and reorder attacks on anchored logs, and can it locate the first altered entry?
2. What is the cost (gas) versus delay trade-off as the anchoring batch size changes?
3. How much latency does logging and anchoring add to each agent step?
4. How does verification time and storage scale from 100 to 100,000 entries?

Scope limits (state these in the thesis)

- The chain proves a log was not changed after it was anchored. It does **not** prove the agent's output was correct or truthful.
- A compromised logger could record false entries before anchoring. Signed entries are an optional

extension.

- Prototype on a test chain, not audited production software.